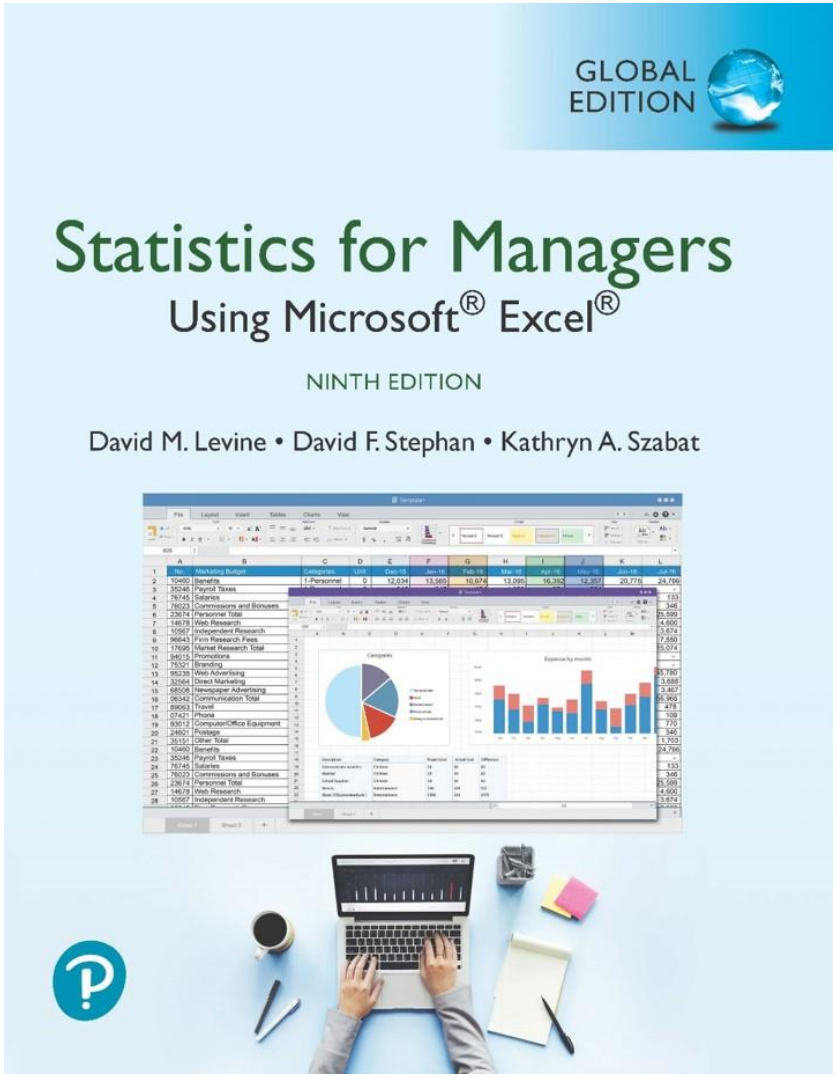


Chapter 14

Introduction to Multiple Regression

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Objectives

In this chapter, you learn:

- How to develop a multiple regression model and interpret the regression coefficients.

- How to determine which independent variables to include in a model and to determine which independent variables are most important in predicting a dependent variable.

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The Multiple Regression Model With k Independent Variables

DCOVA

Idea: Examine the linear relationship between 1 dependent (Y) & 2 or more independent variables (X_i).

Multiple Regression Model with k Independent Variables:

Y-intercept | Population slopes | Random Error

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \cdots + \beta_k X_{ki} + \varepsilon_i$$

Where:

β_0 = Y intercept
β_1 = slope of Y with variable X_1, holding X_2, X_3, X_4, . . . , X_k constant
β_2 = slope of Y with variable X_2, holding X_1, X_3, X_4, . . . , X_k constant
β_3 = slope of Y with variable X_3, holding X_1, X_2, X_4,. . . , X_k constant
.
.
β_k = slope of Y with variable X_k, holding X_1, X_2, X_3,. . . , X_{k-1} constant
ε_i = random error in Y for observation i

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Multiple Regression Model With 2 Independent Variables

DCOVA

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \varepsilon_i$$

Where:

β_0 = Y intercept

β_1 = slope of Y with variable X_1, holding X_2 constant

β_2 = slope of Y with variable X_2, holding X_1 constant

ε_i = random error in Y for observation i

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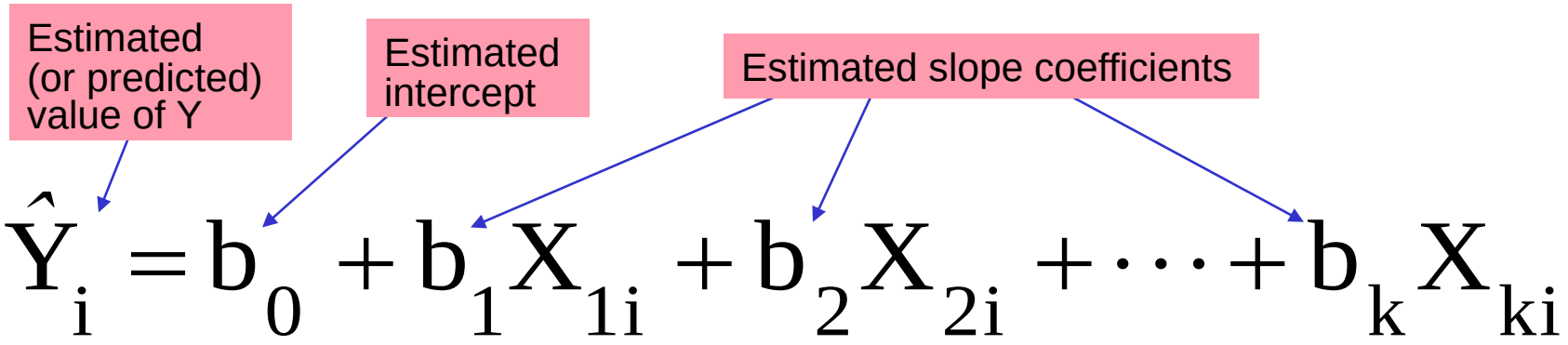
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Multiple Regression Equation

DCOVA

The coefficients of the multiple regression model are estimated using sample data.

Multiple regression equation with k independent variables:



In this chapter we will use Excel to obtain the regression slope coefficients and other regression summary measures.

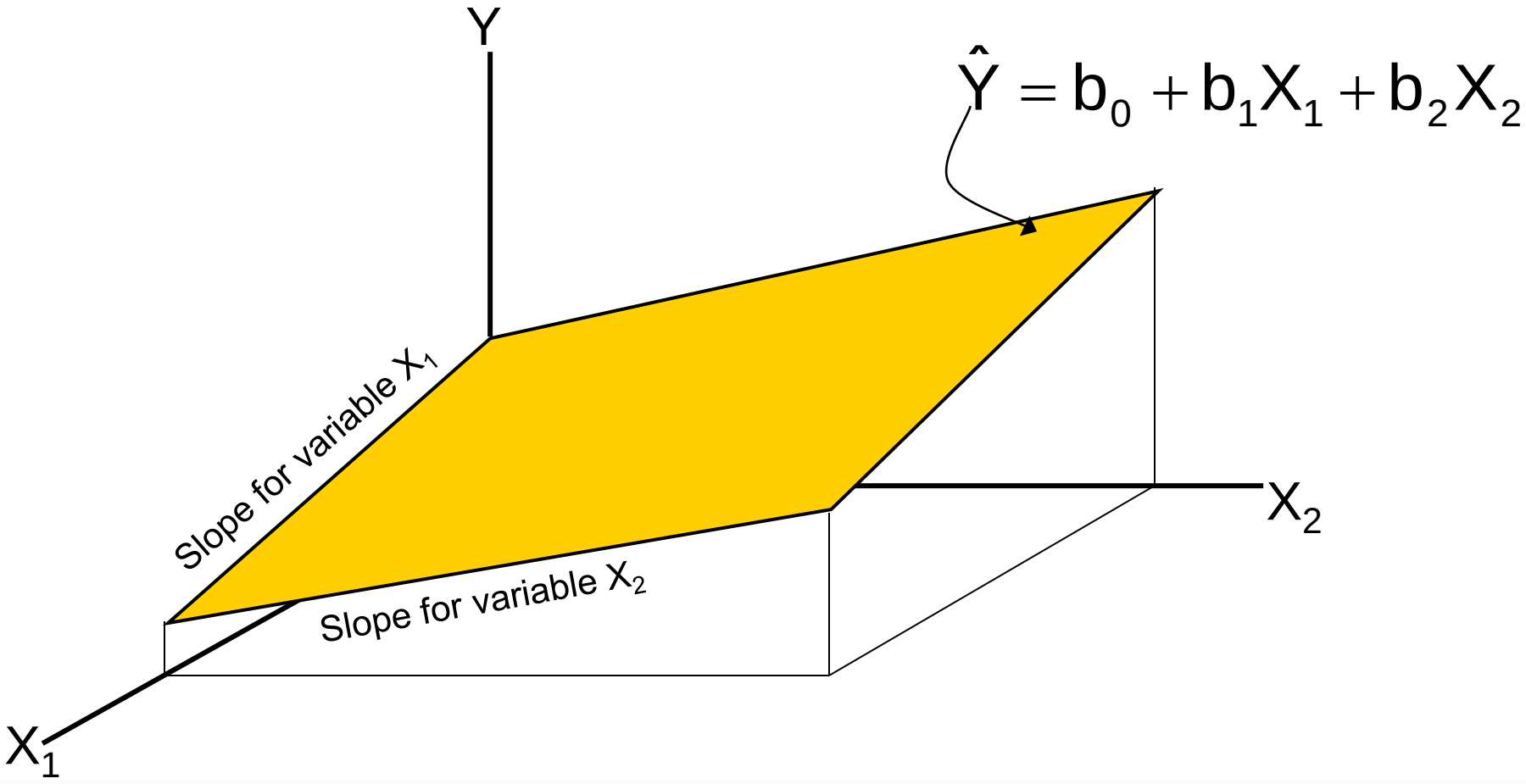
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Multiple Regression Equation With Two Independent Variables

DCOVA

Two variable model



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Example:
2 Independent Variables

DCOVA

- A distributor of frozen dessert pies wants to evaluate factors thought to influence demand.
  - Dependent variable: Pie sales (units per week)
  - Independent variables: Price (in $), Advertising ($100's)
- Data are collected for 15 weeks.

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Pie Sales Example

Week	Pie Sales	Price ($)	Advertising ($100s)
1	350	5.50	3.3
2	460	7.50	3.3
3	350	8.00	3.0
4	430	8.00	4.5
5	350	6.80	3.0
6	380	7.50	4.0
7	430	4.50	3.0
8	470	6.40	3.7
9	450	7.00	3.5
10	490	5.00	4.0
11	340	7.20	3.5
12	300	7.90	3.2
13	440	5.90	4.0
14	450	5.00	3.5
15	300	7.00	2.7

DCOVA

Multiple regression equation:

$$\widehat{Sales} = b_0 + b_1 (Price) + b_2 (Advertising).$$

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Excel Multiple Regression Output

DCOVA

Regression Statistics	
Multiple R	0.72213
R Square	0.52148
Adjusted R Square	0.44172
Standard Error	47.46341
Observations	15

Sales = 306.526 – 24.975(Price) + 74.131(Advertising)

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	2	29460.027	14730.013	6.53861	0.01201
Residual	12	27033.306	2252.776		
Total	14	56493.333			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	306.52619	114.25389	2.68285	0.01993	57.58835	555.46404
Price	-24.97509	10.83213	-2.30565	0.03979	-48.57626	-1.37392
Advertising	74.13096	25.96732	2.85478	0.01449	17.55303	130.70888

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The Multiple Regression Equation

DCOVA

$$\widehat{Sales} = 306.526 - 24.975(Price) + 74.131(Advertising)$$

Where:

Sales is in number of pies per week.
Price is in $.
Advertising is in $100's.

$b_1 = -24.975$: Sales will decrease, on average, by 24.975 pies per week for each $1 increase in selling price, net of the effects of changes due to advertising.

$b_2 = 74.131$: Sales will increase, on average, by 74.131 pies per week for each $100 increase in advertising, net of the effects of changes due to price

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Using The Regression Equation to Make Predictions

DCOVA

Predict sales for a week in which the selling price is $5.50 and advertising is $350:

$$\widehat{\text{Sales}} = 306.526 - 24.975(\text{Price}) + 74.131(\text{Advertising})$$

$$= 306.526 - 24.975\,(5.50) + 74.131\,(3.5)$$

$$= 428.6216$$

Predicted sales is 428.6216 pies.

Note that Advertising is in $100s, so $350 means that $X_2 = 3.5$.

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The Coefficient of Multiple Determination, r^2

DCOVA

- Reports the proportion of total variation in Y explained by the variation of all X variables taken together.

$$r^2 = \frac{SSR}{SST} = \frac{\text{regression sum of squares}}{\text{total sum of squares}}$$

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Multiple Coefficient of Determination In Excel

DCOVA

Regression Statistics	
Multiple R	0.72213
R Square	0.52148
Adjusted R Square	0.44172
Standard Error	47.46341
Observations	15

$$r^2 = \frac{SSR}{SST} = \frac{29{,}460.027}{56{,}493.306} = .52148$$

52.1% of the variation in pie sales is explained by the variation in price and advertising.

ANOVA	*df*	*SS*	*MS*	*F*	*Significance F*
Regression	2	29460.027	14730.013	6.53861	0.01201
Residual	12	27033.306	2252.776		
Total	14	56493.333			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	306.52619	114.25389	2.68285	0.01993	57.58835	555.46404
Price	-24.97509	10.83213	-2.30565	0.03979	-48.57626	-1.37392
Advertising	74.13096	25.96732	2.85478	0.01449	17.55303	130.70888

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Adjusted r^2

(continued)

DCOVA

- Shows the proportion of variation in Y explained by all X variables adjusted for the number of X variables used:

$$r^2_{adj} = 1 - \left[(1 - r^2) \left(\frac{n-1}{n-k-1} \right) \right]$$

(where n = sample size, k = number of independent variables)

- Penalises excessive use of unimportant independent variables.
- Smaller than r^2.
- Useful in comparing among models.

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Is the Overall Model Significant?

DCOVA

- F Test for Overall Significance of the Model.
- Shows if there is a linear relationship between all of the X variables considered together and Y.
- Use F-test statistic.
- Hypotheses:

H_0: $\beta_1 = \beta_2 = \cdots = \beta_k = 0$ (no linear relationship)

H_1: at least one $\beta_i \neq 0$ (at least one independent variable affects Y)

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F Test for Overall Significance

DCOVA

- Test statistic:

$$F_{STAT} = \frac{MSR}{MSE} = \frac{\frac{SSR}{k}}{\frac{SSE}{n-k-1}}$$

where F_{STAT} has numerator d.f.= k and

denominator d.f.= (n – k - 1)

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F Test for Overall Significance

(continued)

DCOVA

H_0: $\beta_1 = \beta_2 = 0$

H_1: β_1 and β_2 not both zero

$\alpha = .05$

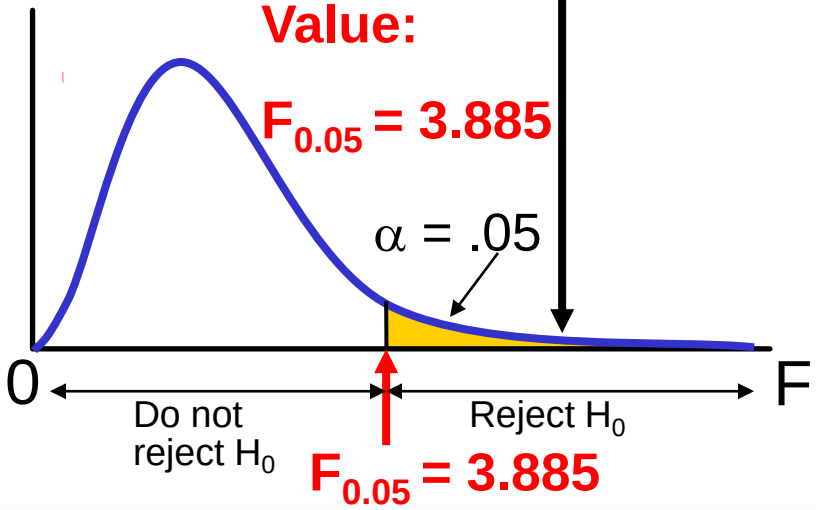
$df_1 = 2$ $df_2 = 12$

Test Statistic:

$$F_{STAT} = \frac{MSR}{MSE} = 6.5386$$

Critical Value:

$F_{0.05} = 3.885$



Decision:

Since F_{STAT} test statistic is in the rejection region (p-value < .05), reject H_0.

Conclusion:

There is evidence that at least one independent variable affects Y.

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Using P-value for F Test

ANOVA	df	SS	MS	F	Significance F
Regression	2	29460.027	14730.013	6.53861	0.01201
Residual	12	27033.306	2252.776		
Total	14	56493.333			

Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	2	29460.027	14730.0	6.5386
Error	12	27033.306	2252.8	Prob > F
C. Total	14	56493.333		0.0120*

Analysis of Variance

Source	DF	SS	MS	F	P
Regression	2	29460	14730	6.54	0.012
Residual Error	12	27033	2253		
Total	14	56493			

Since the P-value = 0.0120 < $\alpha = 0.05$ the null is rejected and conclude the overall model is significant.

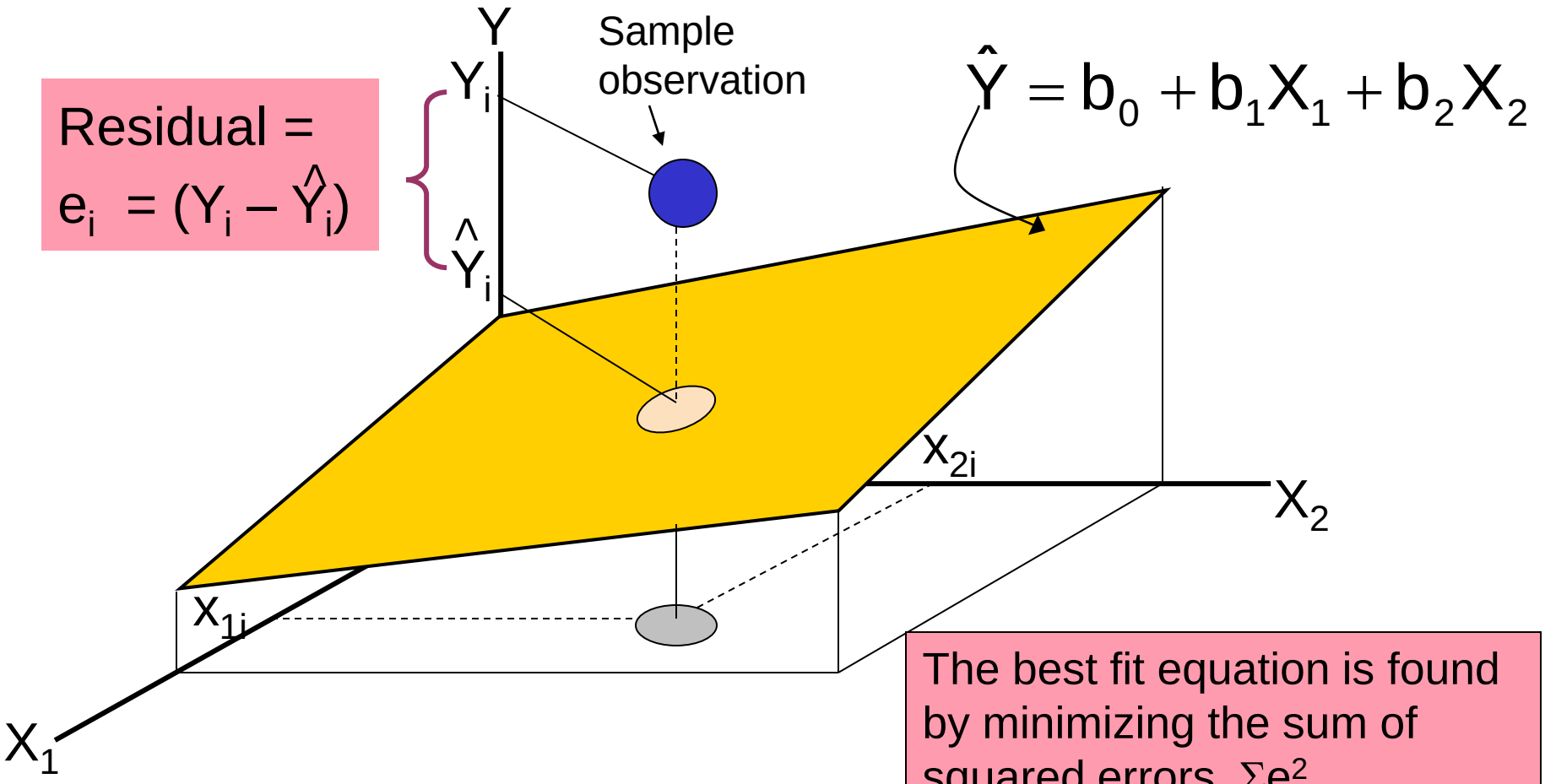
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Residuals in Multiple Regression

DCOVA

Two variable model

Residual = $e_i = (Y_i - \hat{Y}_i)$



The best fit equation is found by minimizing the sum of squared errors, Σe^2.

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Multiple Regression Assumptions

DCOVA

Errors (residuals) from the regression model:

$$e_i = (Y_i - \hat{Y}_i)$$

Assumptions:

- The errors are normally distributed.
- Errors have a constant variance.
- The model errors are independent.

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Residual Plots Used in Multiple Regression to Check Assumptions

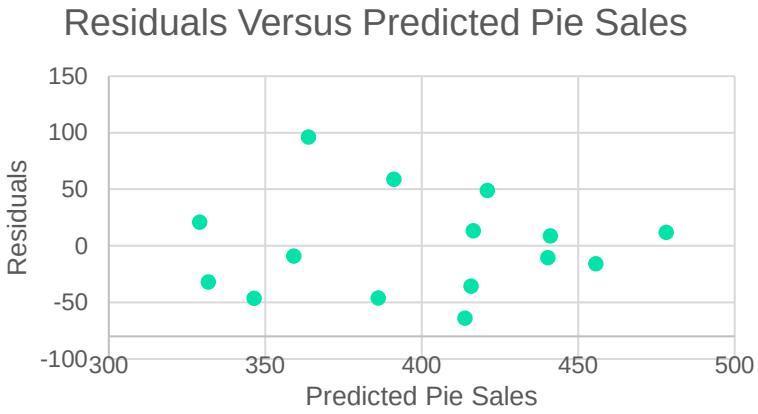
DCOVA

Residual Plot	Significance of Discovered Pattern
Residuals versus the predicted value of Y	1. Evidence of a possible curvilinear effect in at least one independent variable. 2. A possible violation of the equal variance assumption. 3. The need to transform the Y variable.
Residuals versus an independent X variable	1. Suggests evidence of a curvilinear effect indicating the need to add a curvilinear independent variable to the model.
Residuals versus data collected in time order	1. Evidence that the independence of errors assumption has been violated.

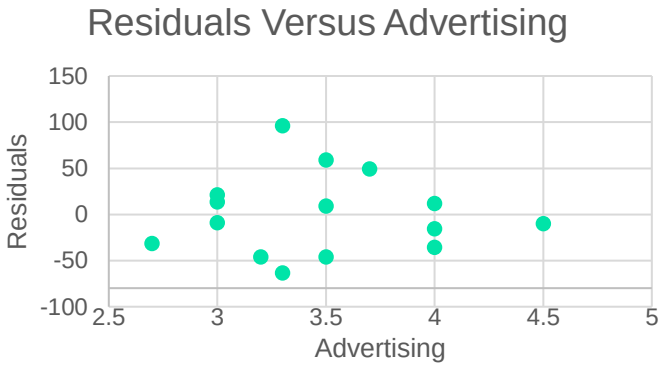
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Residual Plots For The Pie Sales Model

DCOVA







All these plots show little or no pattern so we can conclude the multiple regression model is appropriate for predicting pie sales.

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Are Individual Variables Significant?

DCOVA

- Use t tests of individual variable slopes.
- Shows if there is a linear relationship between the variable X_j and Y holding constant the effects of other X variables.
- Hypotheses:
  - H_0: $\beta_j = 0$ (no linear relationship)
  - H_1: $\beta_j \neq 0$ (linear relationship does exist between X_j and Y)

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Are Individual Variables Significant?

(continued)

DCOVA

H_0: $\beta_j = 0$ (no linear relationship between X_j and Y)

H_1: $\beta_j \neq 0$ (linear relationship does exist between X_j and Y)

Test Statistic:

$$t_{STAT} = \frac{b_j - 0}{S_{b_j}}$$

(df = n – k – 1)

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Inferences about the Slope: t Test Example

DCOVA

$H_0: \beta_j = 0$

$H_1: \beta_j \neq 0$

d.f. = 15-2-1 = 12

α = .05

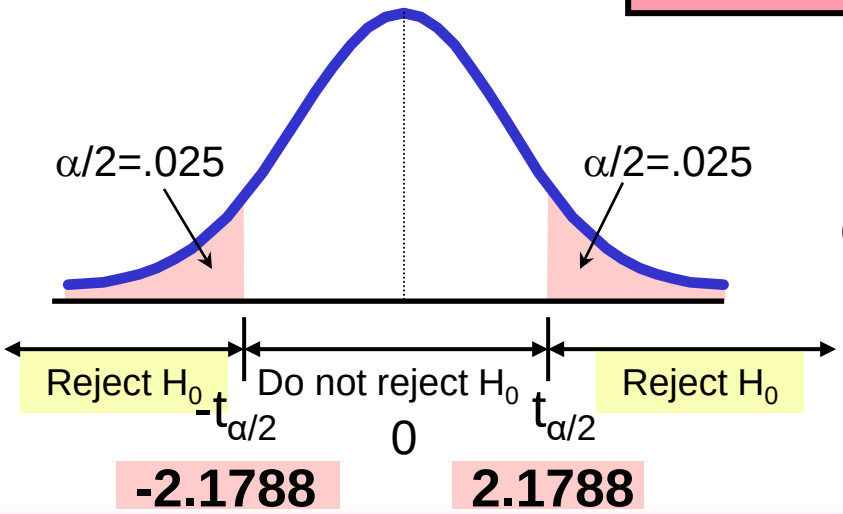
$t_{\alpha/2}$ = 2.1788

From the Excel output:

For Price t_{STAT} = -2.306, with p-value .0398.

For Advertising t_{STAT} = 2.855, with p-value .0145

The test statistic for each variable falls in the rejection region (p-values < .05).



Decision:

Reject H_0 for each variable.

Conclusion:

There is evidence that both Price and Advertising affect pie sales at α = .05.

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Confidence Interval Estimate for the Slope

DCOVA

Confidence interval for the population slope β_j

$$b_j \pm t_{\alpha/2} S_{b_j}$$

where t has
(n – k – 1) d.f.

Here, t has

(15 – 2 – 1) = 12 d.f.

Example: Form a 95% confidence interval for the effect of changes in price (X_1) on pie sales:

-24.975 ± (2.1788)(10.832)

So the interval is (-48.576, -1.374)

(This interval does not contain zero, so price has a significant effect on sales).

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Using Dummy Variables

DCOVA

- A dummy variable is a categorical independent variable with two levels:
  - yes or no, on or off, male or female.
  - coded as 0 or 1.
- Assumes the slopes associated with numerical independent variables do not change with the value for the categorical variable.
- If more than two levels, the number of dummy variables needed is (number of levels - 1).

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Dummy-Variable Example (with 2 Levels)

DCOVA

$$\hat{Y} = b_0 + b_1 X_1 + b_2 X_2$$

Let:

Y = pie sales

X_1 = price

X_2 = holiday (X_2 = 1 if a holiday occurred during the week).
(X_2 = 0 if there was no holiday that week).

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Dummy-Variable Example (with 2 Levels)

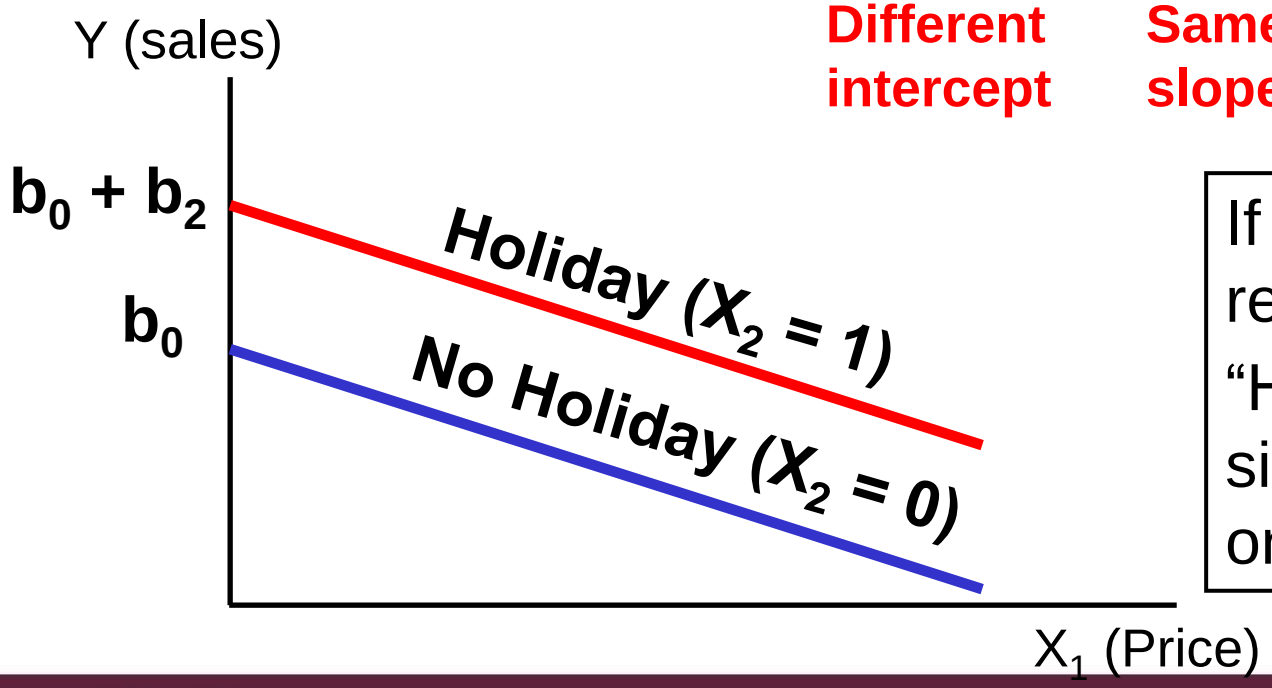
(continued)

DCOVA

$$\hat{Y} = b_0 + b_1X_1 + b_2(1) = (b_0 + b_2) + b_1X_1 \quad \textbf{Holiday}$$

$$\hat{Y} = b_0 + b_1X_1 + b_2(0) = b_0 + b_1X_1 \quad \textbf{No Holiday}$$

Different intercept **Same slope**



If $H_0: \beta_2 = 0$ is rejected, then "Holiday" has a significant effect on pie sales.

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Dummy-Variable Example Excel Output *(continued)*

DCOVA

Regression Statistics	
Multiple R	0.785273784
R Square	0.616654916
Adjusted R Sq	0.552764069
Standard Error	42.4818016
Observations	15

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	2	34836.89173	17418.45	9.651694	0.003173506
Residual	12	21656.4416	1804.703		
Total	14	56493.33333			

	Coefficients	*Standard Error*	*t Stat*	*P-value*	*Lower 95%*	*Upper 95%*
Intercept	548.0984369	65.08044138	8.421861	2.21E-06	406.3003363	689.8965376
Price	-26.12839289	9.707921784	-2.69145	0.019617	-47.2801374	-4.97664836
Holiday	90.11500528	24.84803909	3.626645	0.003472	35.97577893	144.2542316

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Interpreting the Dummy Variable Coefficient (with 2 Levels)

DCOVA

$$\hat{Sales} = 548.1 - 26.1(Price) + 90.1(Holiday)$$

Sales: number of pies sold per week

Price: pie price in $

Holiday: $\begin{cases} 1 & \text{If a holiday occurred during the week} \\ 0 & \text{If no holiday occurred} \end{cases}$

b_2 = 90.1: on average, sales were 90.1 pies greater in weeks with a holiday than in weeks without a holiday, given the same price.

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Dummy-Variable Models (more than 2 Levels)

DCOVA

- The number of dummy variables is **one less than the number of levels.**

- Example:

  Y = house price ; X_1 = square feet.

- If style of the house is also thought to matter:

  Style = double story, farm, other.

Three levels, so two dummy variables are needed.

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Dummy-Variable Models (more than 2 Levels)

(continued)

DCOVA

- Example: Let "other" be the default category, and let X_2 and X_3 be used for the other two categories:

Y = house price

X_1 = square feet

X_2 = 1 if farm, 0 otherwise

X_3 = 1 if double story, 0 otherwise

The multiple regression equation is:

$$\hat{Y} = b_0 + b_1X_1 + b_2X_2 + b_3X_3$$

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Interpreting the Dummy Variable Coefficients (with 3 Levels)

DCOVA

Consider the regression equation:

$$\hat{Y} = 20.43 + 0.045X_1 + 23.53X_2 + 18.84X_3$$

For style = other: $X_2 = X_3 = 0$

$$\hat{Y} = 20.43 + 0.045X_1$$

For a farm: $X_2 = 1; X_3 = 0$

$$\hat{Y} = 20.43 + 0.045X_1 + 23.53$$

With the same square feet, a farm will have an estimated average price of 23.53 thousand dollars more than an other style.

For a double story: $X_2 = 0; X_3 = 1$

$$\hat{Y} = 20.43 + 0.045X_1 + 18.84$$

With the same square feet, a double story will have an estimated average price of 18.84 thousand dollars more than an other style.

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Cross-Validation is a process for Testing Predictive Models on "Hold Out Data"

DCOVA

- **Cross-Validation** helps prevents **Overfitting**, a model that fits a data set too-well (fitting quirks and randomness specific to only that specific set of data).
- Overfitting is often a problem when using a model for predicting **future data**.
- **Cross-validation** is used to create better-performing predictive models.
- Cross-validation techniques split data via different methods to enable model(s) to be tested on "hold out data" (data that isn't used to build the model(s).

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Examples of How To "Hold Out Data"

DCOVA

- Randomly split original data into 2 groups (Training and Test).
- Randomly split original data into 3 groups (Training, Validation, Test).
- k-fold cross-validation randomly splits the data into k equal parts (sometimes called folds) and runs k regression analyses in which a different fold is held out each time.
- LOOCV (leave one out cross validation) is used for small data sets where points are left out one at a time, so k = n.

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Chapter Summary

In this chapter we discussed:

- How to develop a multiple regression model and interpret the regression coefficients.

- How to determine which independent variables to include in a model and to determine which independent variables are most important in predicting a dependent variable.

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